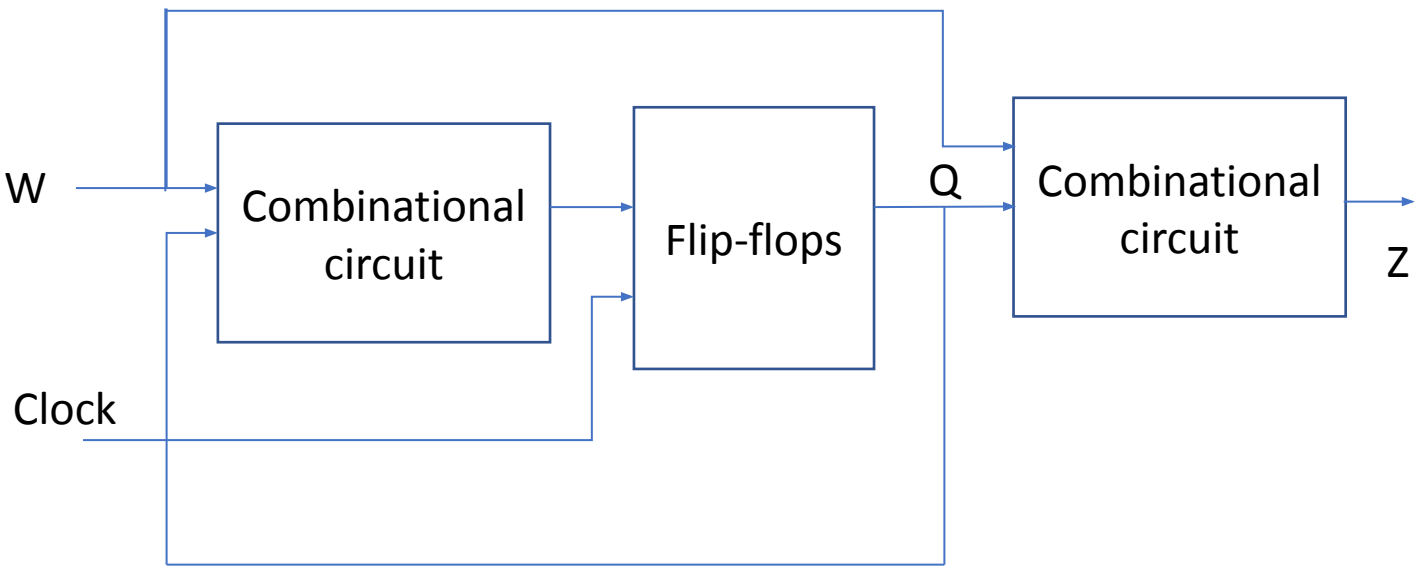
CSE 460: VLSI Design (Lab)

Experiment 4

Simulation of Mealy and Moore Type FSMs in Verilog

Finite State Machine

- In Combinational logic circuits, outputs are determined solely by the present values of the inputs.
- In **Sequential circuits, outputs depend both on the past behavior of the circuit and the present values of the inputs**. The storage elements in terms of flip-flops can reserve the state information of the logic circuit at any given time.
- The sequential circuits in which a clock signal is used to control the operation of the circuit, is called synchronous sequential circuits. These synchronous sequential circuits are known in general as Finite State Machines (FSMs).
- The **building blocks of a finite state machine are combinational circuits and one or more flip-flops**. Under the control of the clock signal, the flip-flops change their state as determined by the combinational logic.



Finite State Machine

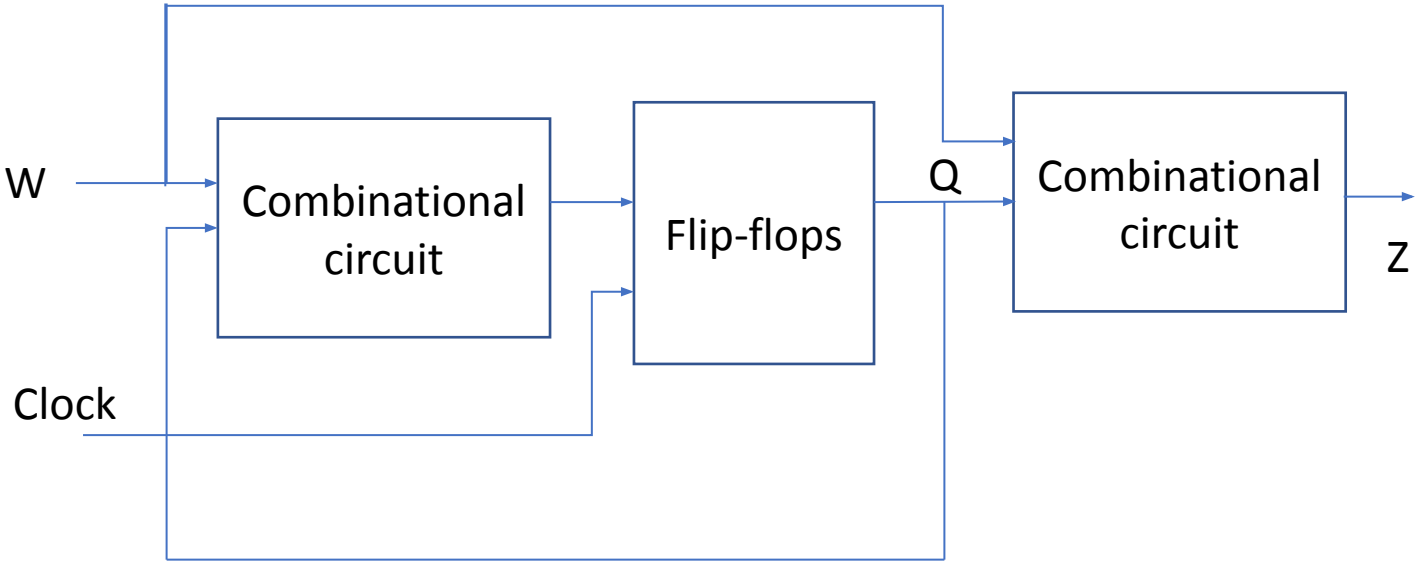
FSMs can be of two types:

1. Moore type FSM

   The sequential circuits whose outputs depend only on the states of the circuit are of Moore type.

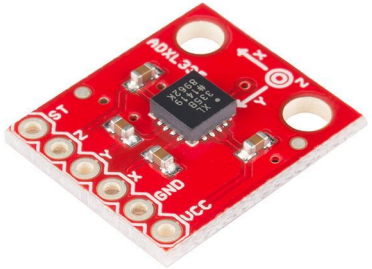
2. Mealy type FSM

   The sequential circuits whose outputs depend on both the state and the present primary inputs are of Mealy type



FSM based motor speed controller

Accelerometer



Motor speed (rpm)

Comparator

Compares the accelerometer reading with the preset reference speed and outputs $\boldsymbol{w} = \mathbf{1}$ if the motor speed exceeds the reference value

$\mathbf{500}\ \boldsymbol{rpm}$

$\boldsymbol{w}$

Finite State Machine (FSM)

Based on the comparator output the FSM will enable the speed controller if $\mathbf{w} = \mathbf{1}$ for two consecutive clock cycles

$\boldsymbol{z}$

Speed Controller

Reduces the motor speed if the output from the FSM is high i.e. $\boldsymbol{z} = \mathbf{1}$

DC motor

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Moore Type FSM

Problem:

Suppose that we wish to design a circuit that meets the following specification:

- The circuit has one input, **w**, and one output, **z**.
- All changes in the circuit occur on the positive edge of a clock signal.
- The output **z** is equal to **1** if during **two immediately preceding** clock cycles the input **w** was equal to 1. Otherwise, the value of z is equal to 0.

Input-output combination:

Clock Cycle	1	2	3	4	5	6	7	8	9	10
w	0	1	0	1	1	0	1	1	1	0
z	0	0	0	0	0	1	0	0	1	1

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Moore Type FSM

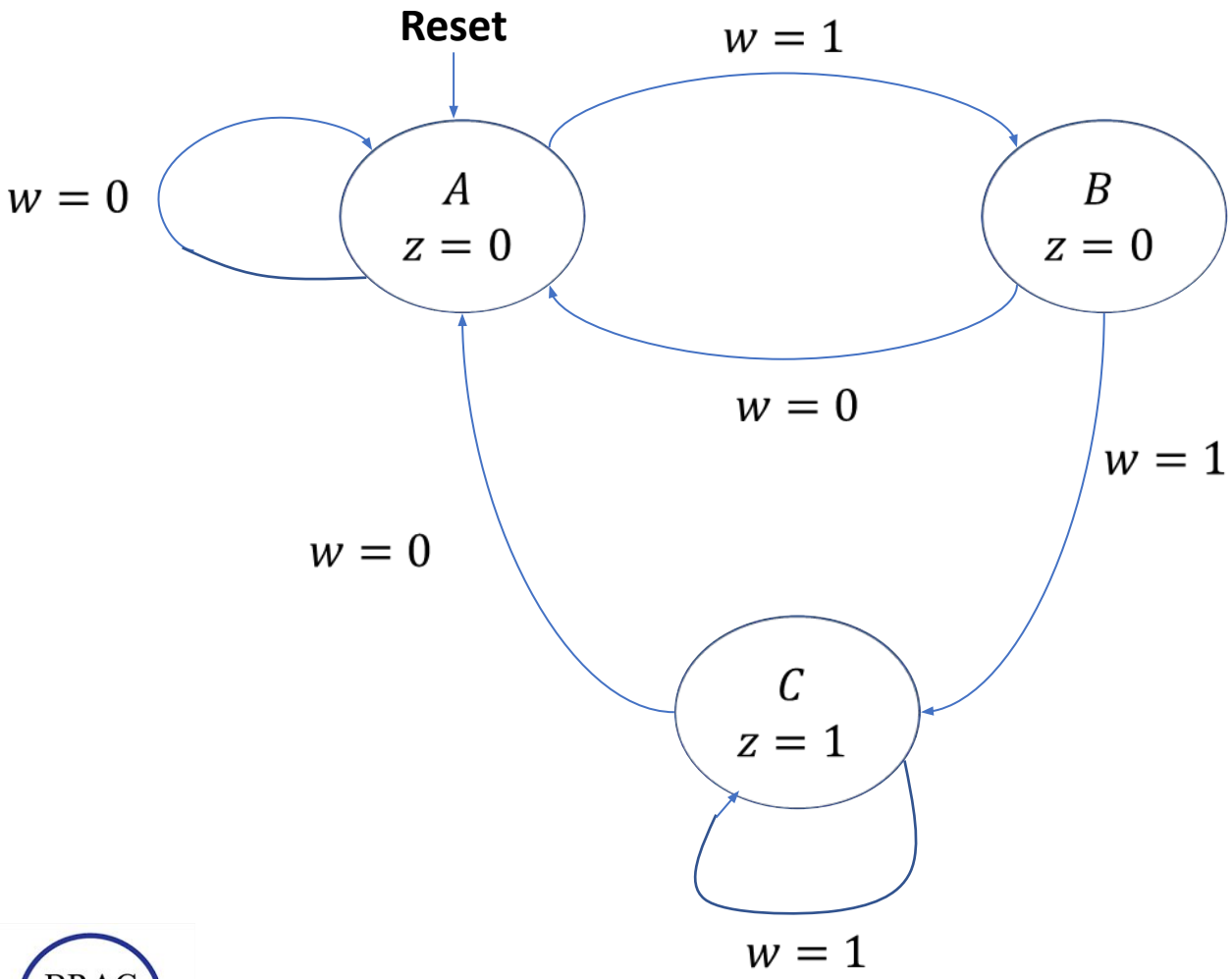


Figure : State Diagram

Present State	Next State		Output z
	w=0	w=1	
A	A	B	0
B	A	C	0
C	A	C	1

Figure : State Table

Moore Type FSM

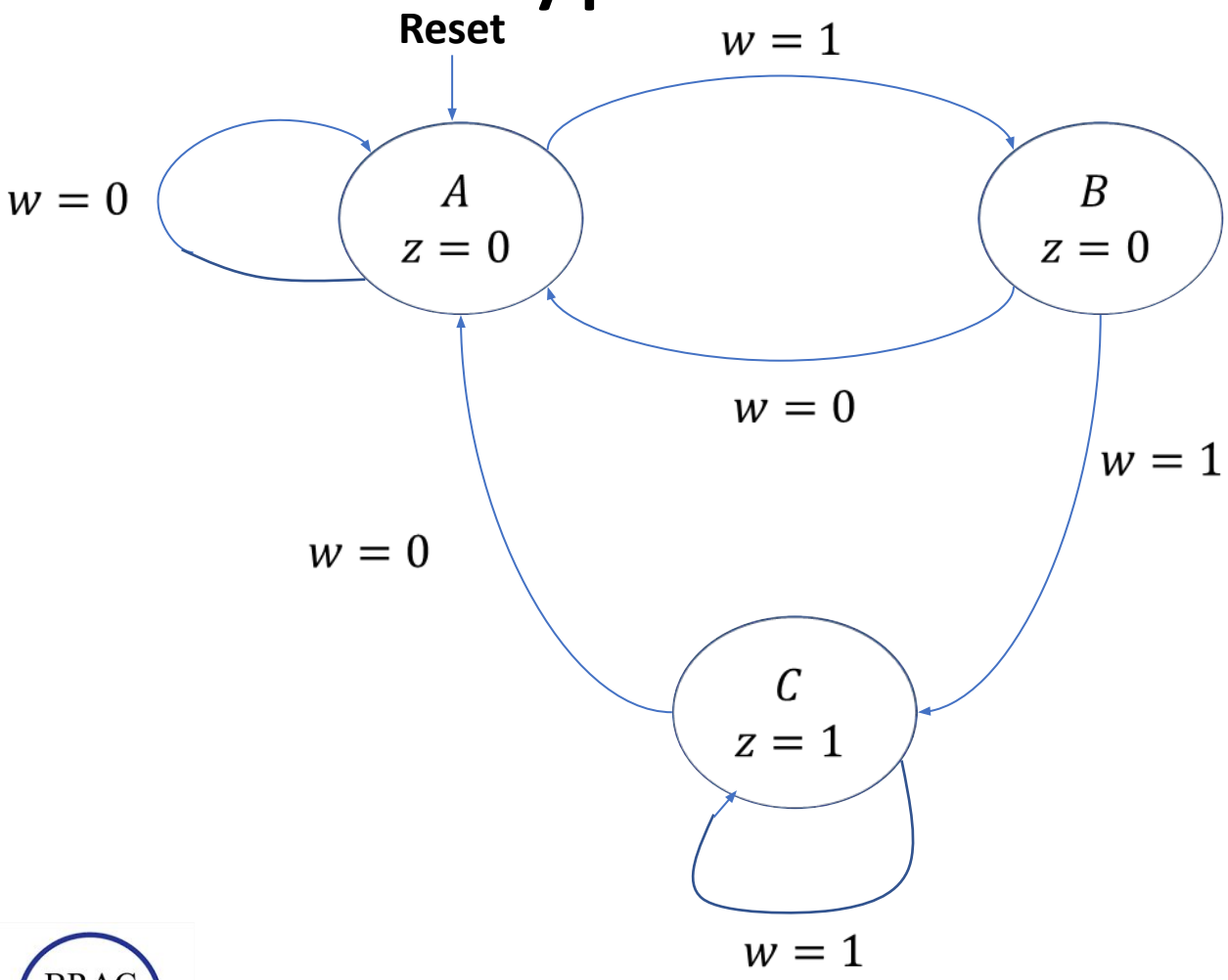
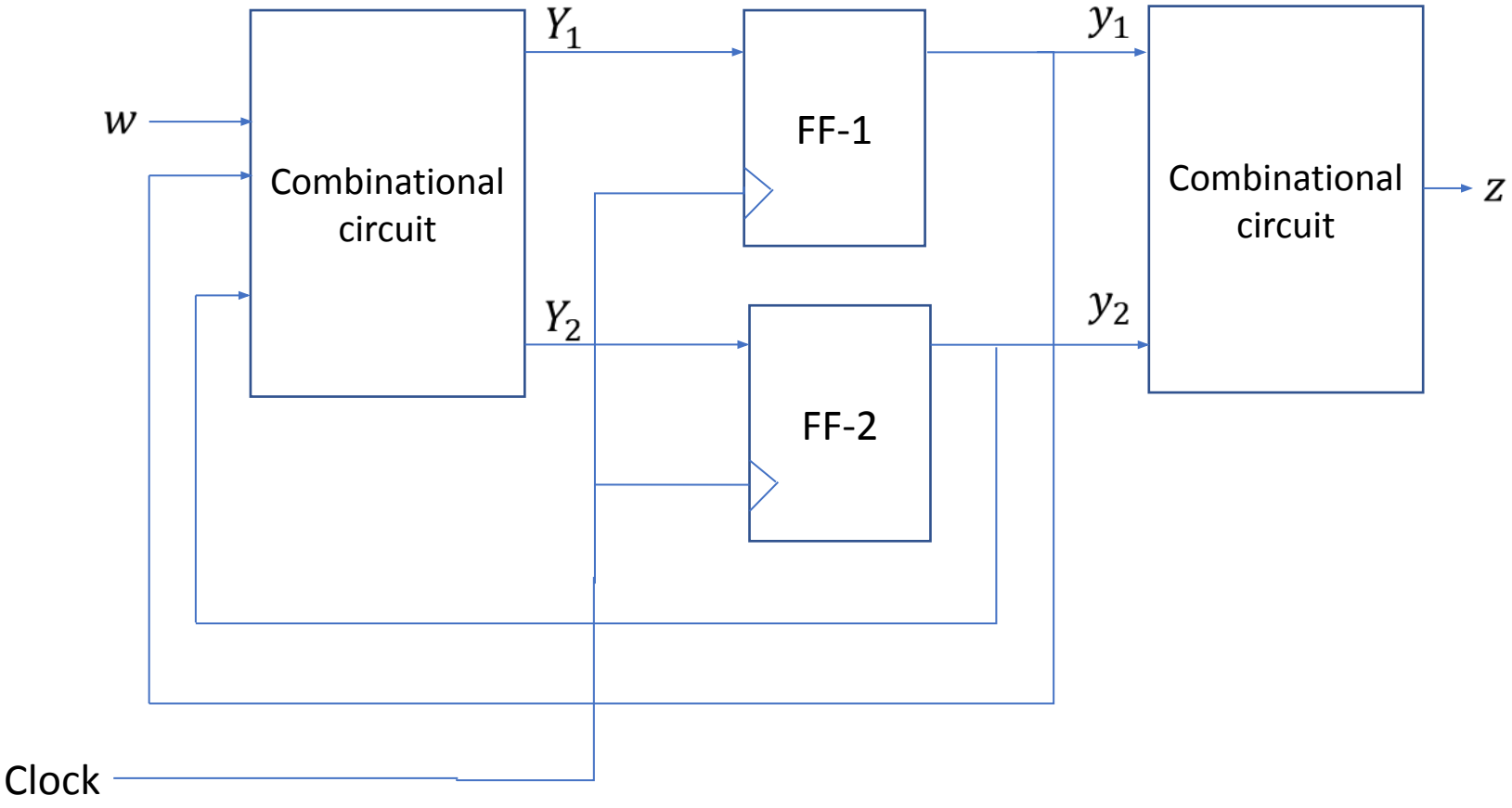


Figure : State Diagram

	Next State		Output
	w=0	w=1	z
00(A)	00(A)	01(B)	0
01(B)	00(A)	10(C)	0
10(C)	00(A)	10(C)	1
11	dd	dd	d

General Circuit Diagram of the FSM

	Next State		Output
	w=0	w=1	z
00	00	01	0
01	00	10	0
10	00	10	1
11	dd	dd	d



Designing the FSM in Verilog

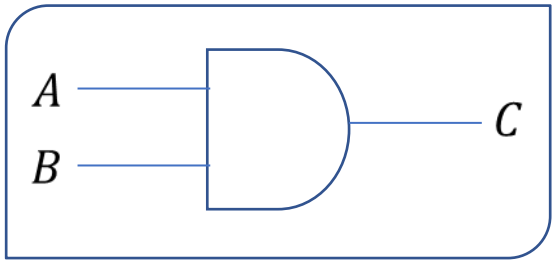
In order to design the FSM using Verilog we will need to perform the following steps:

- Assigning binary variables to states
- A mechanism for figuring out the current state (Sequential circuit block)
- Defining the logic for state transition (Combinational circuit block)
- Defining the output for the current state (Continuous assignment statement)

Combinational circuit synthesis in Verilog

`always@(*)` blocks are used to describe combinational logic in Verilog. *Blocking assignment* statements are to be used inside these blocks.

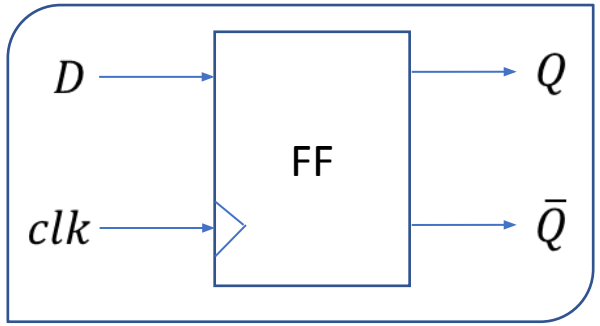
```
always@(*)
begin
    C = A & B;
end
```



Sequential circuit synthesis in Verilog

`always@(posedge clk)` or `always@(negedge clk)` blocks are used to describe combinational logic in Verilog. *Non-blocking assignment* statements are to be used inside these blocks.

```
always@(posedge clk)
begin
    Q <= D;
end
```



Moore Type FSM

	Next State		Output
			z
	w=0	w=1	
00(A)	00(A)	01(B)	0
01(B)	00(A)	10(C)	0
10(C)	00(A)	10(C)	1
11	dd	dd	d

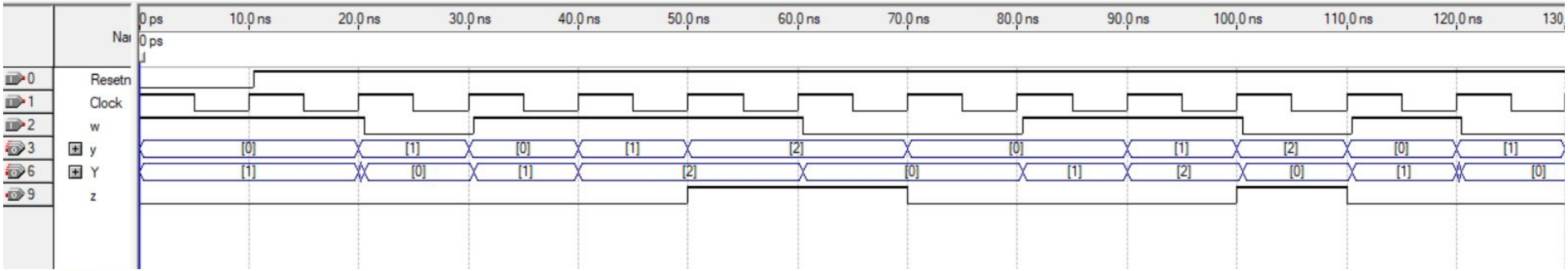
```
module Moore11 (Clock, Resetn, w, z,y,Y);

input Clock, Resetn, w;
output reg z;
output reg [2:1] y, Y;
parameter [2:1] A = 2'b00, B = 2'b01, C = 2'b10;


        // Define the next state combinational circuit
        always @(w, y)

                case (y)

                A: if (w) Y = B;
                        else Y = A;
                B: if (w) Y = C;
                        else Y = A;
                C: if (w) Y = C;
                        else Y = A;
                default: Y = 2'bxx;

        endcase


        // Define the sequential block
        always @(negedge Resetn, posedge Clock) begin

        if (Resetn == 0) y <= A;

        else y <= Y;

        end

        // Define output

        always@(y) begin

        case (y)

                A: z=0;
                B: z=0;
                C: z=1;


        endcase

        end


endmodule
```

Moore Type FSM

Steps for properly inserting inputs (w as well as resetn):

1. first invert the Clock signal
2. Introduce now a small offset(0.5ns) in both w and Resetn signals

Video link : **https://youtu.be/ZdOeTZLUr0Q?feature=shared**



Mealy Type FSM

Problem:

We wish to design a circuit that meets the following specification:

- The circuit has one input, **w**, and one output, **z**.
- All changes in the circuit occur on the positive edge of a clock signal.
- The output **z** is equal to **1** if during **two immediate** clock cycles the input **w** was equal to 1. Otherwise, the value of z is equal to 0.

Input-output combination:

Clock Cycle	1	2	3	4	5	6	7	8	9	10
w	0	1	0	1	1	0	1	1	1	0
z	0	0	0	0	1	0	0	1	1	0

Mealy Type FSM

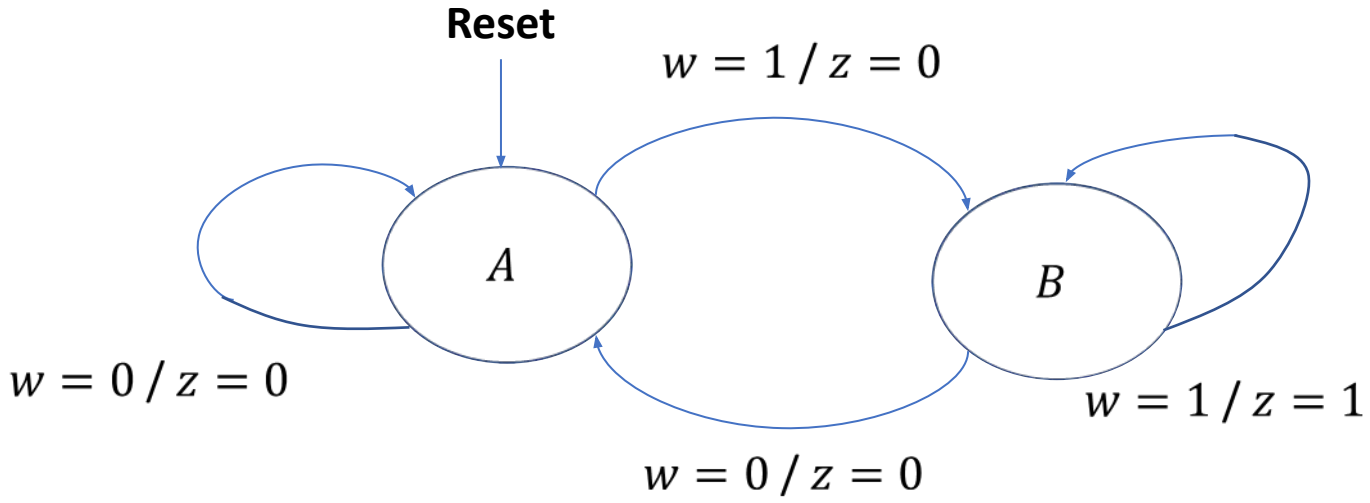


Figure : State Diagram

Present State	Next State		Output z	
	w=0	w=1	w=0	w=1
A	A	B	0	0
B	A	B	0	1

Figure : State Table

Mealy Type FSM

Present State	Next State		Output z	
	w=0	w=1	w=0	w=1
A	A	B	0	0
B	A	B	0	1

Figure : State Table

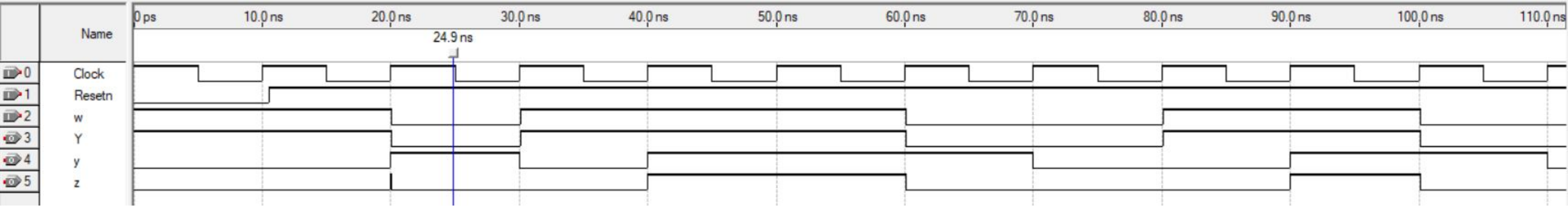
Present State y	Next State Y		Output z	
	w=0	w=1	w=0	w=1
0(A)	0(A)	1(B)	0	0
1(B)	0(A)	1(B)	0	1

Figure : State-assigned Table

```
module Mealy11 (Clock, Resetn, w, z,y,Y);

	input Clock, Resetn, w;
	output reg z;
	output reg y, Y;
	parameter A = 1'b0, B = 1'b1;


	// Define the next state and output "combinational" circuits
	always @(w, y)
	case (y)

		A:	if (w)

			begin
			z = 0;
			Y = B;
			end

			else
			begin
			z = 0;
			Y = A;
			end


		B:	if (w)

			begin
			z = 1;
			Y = B;
			end

			else
			begin
			z = 0;
			Y = A;
			end

	endcase


	// Define the sequential block
	always @(negedge Resetn, posedge Clock) begin
	if (Resetn == 0) y <= A;
	else y <= Y;

	end


endmodule
```

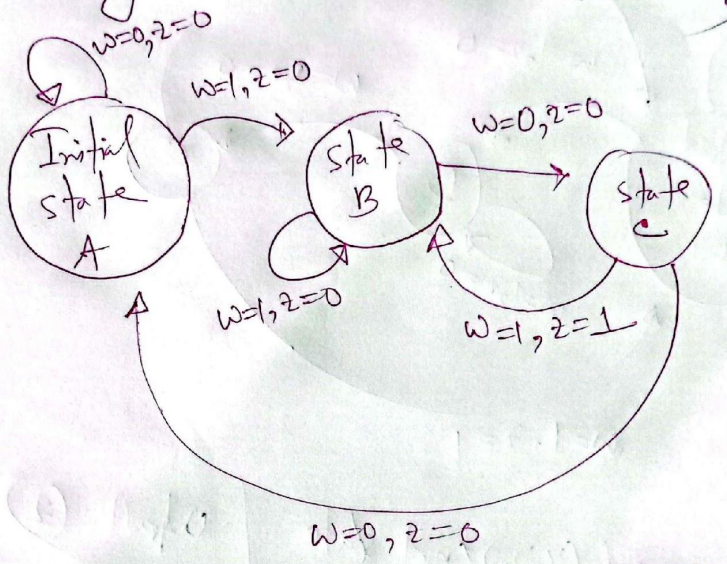
Steps for properly inserting inputs (w as well as resetn):

1. first invert the Clock signal
2. Introduce now a small offset(0.5ns) in both w and Resetn signals



Mealy Type FSM

(+) Detecting Sequence : "101" (Overlapping)

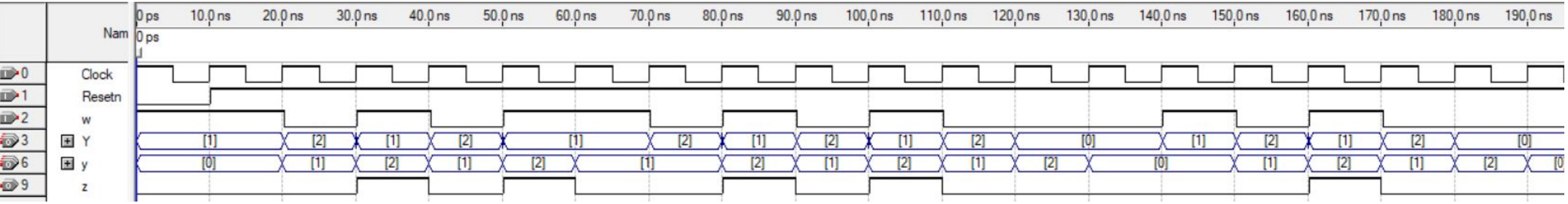


Present State (y_2y_1)	Next State (Y_2Y_1)		Output (z)	
	w=0	w=1	w=0	w=1
(00) A	A(00)	B(01)	0	0
(01) B	C(10)	B(01)	0	0
(10) C	A(00)	B(01)	0	1

```
module Mealy101(Clock, Resetn, w, z,y,Y);

        input Clock, Resetn, w;
        output reg z;
        output reg [1:0]y, Y;
        parameter A = 2'b00, B = 2'b01, C= 2'b10;


        // Define the next state and output "combinational" circuits
        always @(w, y)
        case (y)

                A:      if (w)

                        begin
                        z = 0;
                        Y = B;
                        end

                        else
                        begin
                        z = 0;
                        Y = A;
                        end


                B:      if (w)

                        begin
                        z = 0;
                        Y = B;
                        end

                        else
                        begin
                        z = 0;
                        Y = C;
                        end

                C:      if (w)

                        begin
                        z = 1;
                        Y = B;
                        end

                        else
                        begin
                        z = 0;
                        Y = A;
                        end

        endcase


        // Define the sequential block
        always @(negedge Resetn, posedge Clock) // Asynchronous reset
        begin
        if (Resetn == 0) y <= A;
        else y <= Y;

        end


endmodule
```

Steps for properly inserting inputs (w as well as resetn):

1. first invert the Clock signal
2. Introduce now a small offset(0.5ns) in both w and Resetn signals



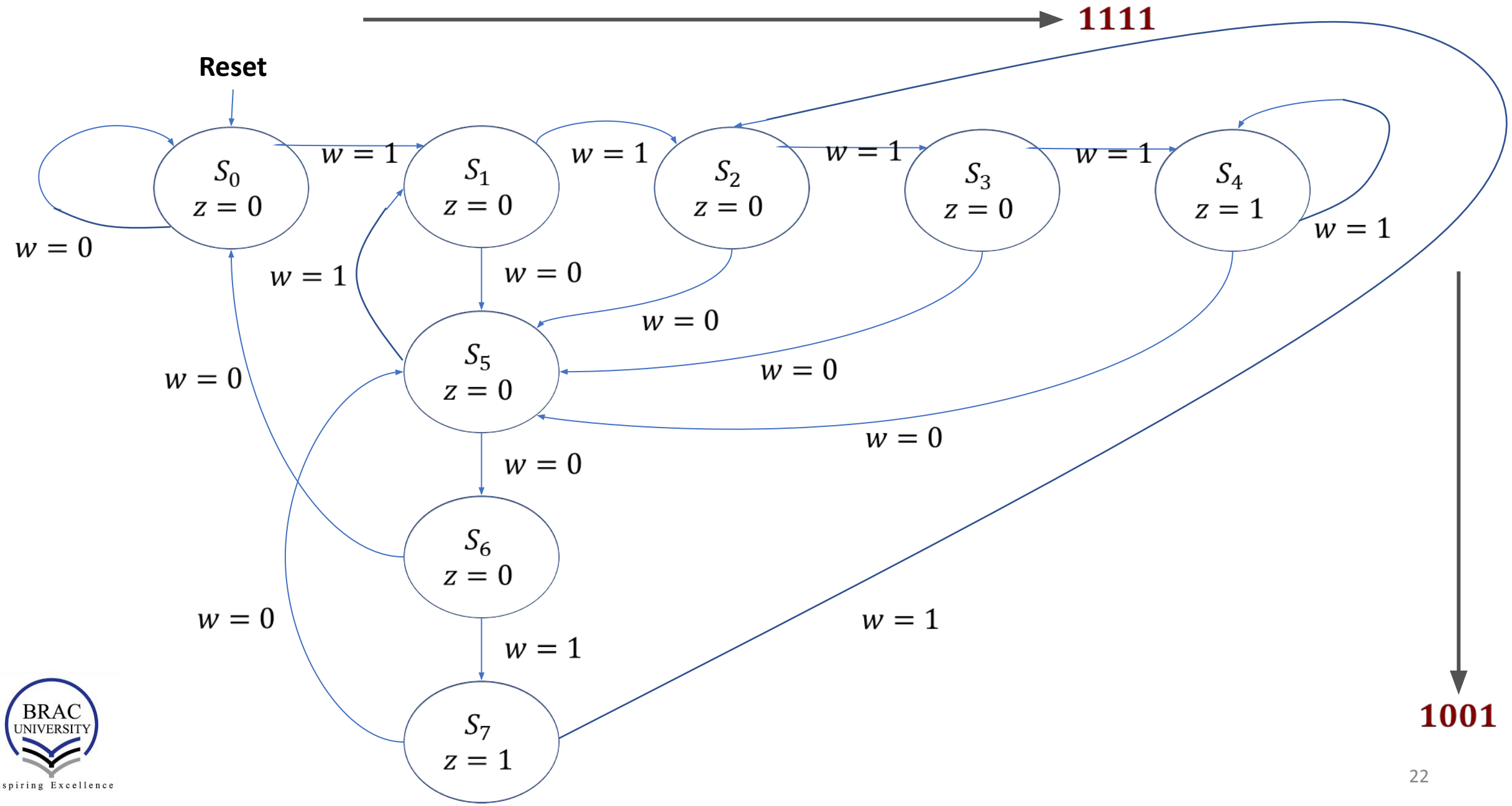
For Practice at home

An Example (Moore type)

Problem: Derive the state diagram for an FSM that has an input w and an output z. The machine has to generate z = 1 when the previous four values of w were 1001 or 1111; otherwise, z = 0.Overlapping input patterns are allowed. An example of the desired behavior is

Clock cycle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
w	0	1	0	1	1	1	1	0	0	1	1	0	0	1	1	1	1
z	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	0	0

State diagram(Moore type)



State table (Moore type)

Present State	Next State w=0	Next State w=1	Output z
S0	S0	S1	0
S1	S5	S2	0
S2	S5	S3	0
S3	S5	S4	0
S4	S5	S4	1
S5	S6	S1	0
S6	S0	S7	0
S7	S5	S2	1

Figure : State Table

	Next State w=0	Next State w=1	Output z
000(S0)	000(S0)	001(S1)	0
001(S1)	101(S5)	010(S2)	0
010(S2)	101(S5)	011(S3)	0
011(S3)	101(S5)	100(S4)	0
100(S4)	101(S5)	100(S4)	1
101(S5)	110(S6)	001(S1)	0
110(S6)	000(S0)	111(S7)	0
111(S7)	101(S5)	010(S2)	1

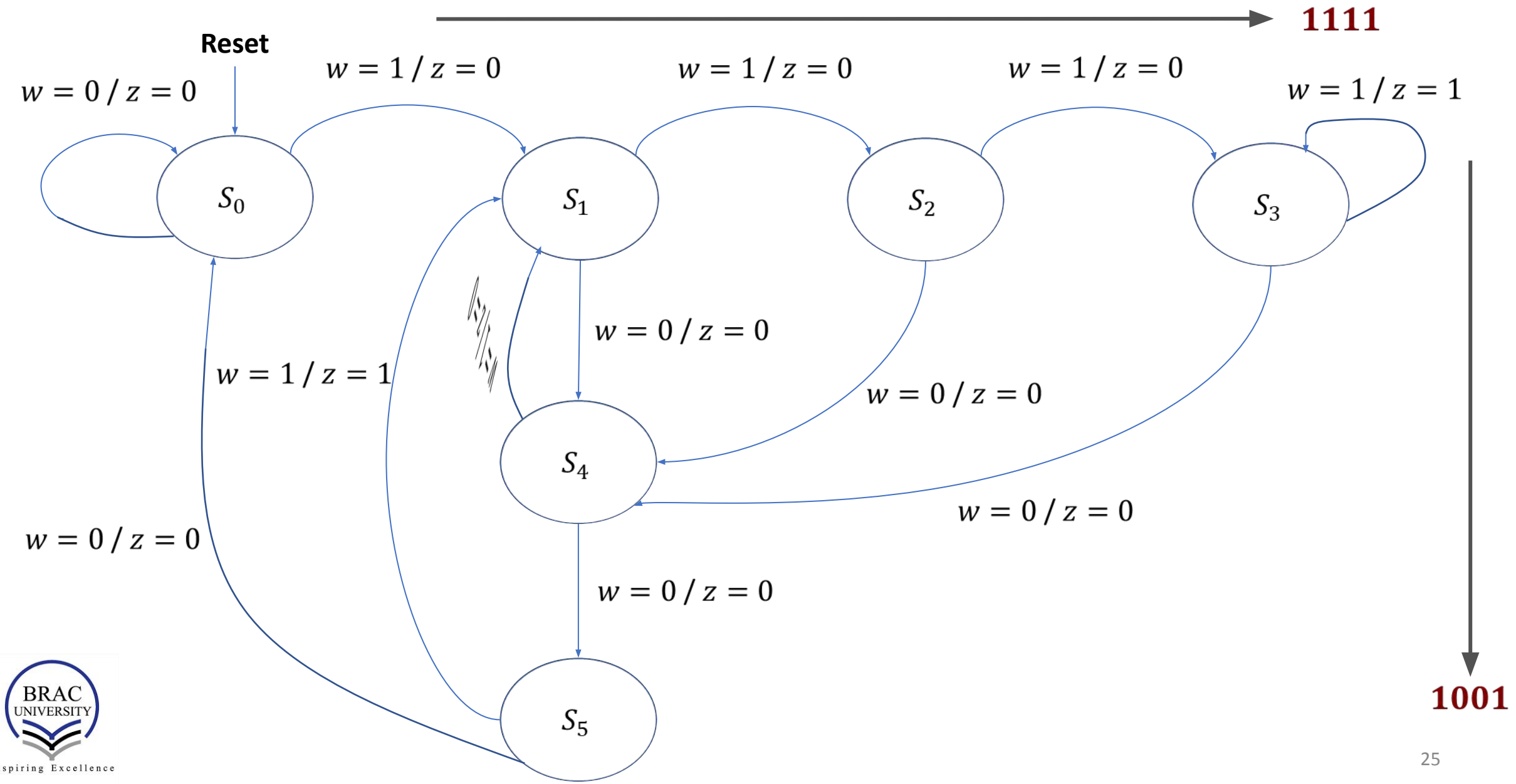
Figure : State-assigned Table

An Example (Mealy type)

- Problem: Derive the state diagram of the Mealy version for the example FSM that is, the FSM will generate $z = 1$ at the instant the input values of w have the pattern 1111 or 1001. Overlapping of the input patterns are allowed.

Clock cycle	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
w	0	1	0	1	1	1	1	0	0	1	1	0	0	1	1	1	1
z	0	0	0	0	0	0	1	0	0	1	0	0	0	1	0	0	1

State diagram(Mealy type)



State table (Mealy type)

Present State	Next State		Output z	
	w=0	w=1	w=0	w=1
S0	S0	S1	0	0
S1	S4	S2	0	0
S2	S4	S3	0	0
S3	S4	S3	0	1
S4	S5	S1	0	0
S5	S0	S1	0	1

Figure : State Table

	Next State		Output z	
	w=0	w=1	w=0	w=1
000(S0)	000(S0)	001(S1)	0	0
001(S1)	100(S4)	010(S2)	0	0
010(S2)	100(S4)	011(S3)	0	0
011(S3)	100(S4)	011(S3)	0	1
100(S4)	101(S5)	001(S1)	0	0
101(S5)	000(S0)	001(S1)	0	1

Figure : State-assigned Table

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